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TOPICAL REVIEW

A Systematic Literature Review on Privacy Preservation in VANETs: Trends, Challenges, and Future Directions

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ABSTRACT Privacy preservation is a fundamental requirement in vehicular ad hoc networks (VANETs) because it addresses sensitive vehicle and driver data. The use of pseudonyms instead of actual vehicle identities is one way to protect vehicle owners from the unauthorized tracking or linking of vehicle movements. In this paper, we conduct a systematic literature review to identify the implementation of pseudonyms in existing privacy preservation schemes via the Kitchenham method. To our knowledge, no study has presented an analysis of the implementation of comprehensive pseudonym life cycles in the last five years. We classify VANET system models and their advancements with the support of emerging technologies, mapping the integration of these technologies to the implementation of the pseudonym life cycle. Furthermore, we categorize the privacy analyses employed in existing schemes. Finally, we identify research gaps and future research directions to advance the development of robust privacy preservation schemes in VANETs. Of the 594 reviewed research papers, 113 satisfied our requirements for primary research (PR). Among them, 71% adopted VANET standard system models, 12% implemented a comprehensive pseudonym life cycle, and 73% conducted both formal and informal security and privacy analyses. We highlight research challenges and suggest future research topics for developing robust privacy-preserving schemes in VANETs.

INDEX TERMS Privacy preservation, comprehensive pseudonym lifecycle, VANET security.

I. INTRODUCTION

Intelligent transport system (ITS) development has advanced significantly in recent decades. By embedding integrated sensors, it is possible to identify hazardous driving circumstances and communicate this information to surrounding vehicles. Vehicular communication is the term used to describe this type of information exchange between moving vehicles. The aim of vehicular communication is to improve driving conditions and make roads safer for everyone.

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A vehicular ad hoc network (VANET) is an ad hoc network involving vehicles. The main goal of adopting a VANET is to reduce the number of collisions by providing drivers access to real-time traffic information so that they can drive safely. While the concept of cooperative intelligent transportation networks is admirable, the inherent openness of VANETs poses substantial hazards to vehicle owners. The continuous sharing of data throughout the network allows attackers to profile vehicles, thereby jeopardizing the privacy of their owners. Data privacy in VANETs is also one of the most important challenges because it supports emergency applications and addresses life-critical information [1]. Many



FIGURE 1. Pseudonym's Lifecycle.

people are reluctant to divulge details about their vehicles or other information that could indicate their intended route.

One way to enable communication between vehicles without disclosing their true identities is through anonymity, which refers to the inability to be uniquely identified within a group [2]. This result can be achieved via pseudonyms, which serve as temporary identifiers that hide the true identities of vehicles and allow their participation in VANETs. However, effective pseudonym management requires a comprehensive pseudonym life cycle, which consists of five key stages: issuance, usage, changing, resolution, and revocation (see Figure 1). Vehicles are assigned cryptographically secure pseudonyms through issuance, and anonymous but verified communication is enabled through usage. Changing strategies change pseudonyms on a regular basis to avoid tracking. Resolution enables authorized entities to identify a pseudonym's true identity in security incident scenarios. Finally, revocation reduces risks and unwanted access by blocking compromised vehicles from connecting to the network. However, the existing implementations have severe drawbacks, owing mostly to the incomplete adoption of the pseudonym life cycle. Our analysis shows that all five steps were implemented in only 14% of existing studies, with most schemes focused solely on issuance and usage while disregarding changing, resolution, and revocation. Ineffective pseudonym-changing strategies, such as fixed-time modifications that ignore vehicular density or communication patterns, result in linkability attacks, in which adversaries correlate previous and new pseudonyms to locate vehicles. Furthermore, the lack of a reliable pseudonym resolution method creates a privacy authority problem, which makes tracking malicious vehicles participating in security incidents more difficult. Finally, the absence of an effective revocation method allows compromised or misbehaving vehicles to continue engaging in the network, thus increasing the risk of Sybil attacks and unauthorized access.

Numerous VANET studies have contributed to the understanding of privacy, particularly in relation to the life cycle of pseudonyms. With respect to pseudonymity approaches, Petit et al. [3] reviewed the literature on pseudonymity in VANETs and classified approaches according to the types of cryptographic techniques used, such as symmetric, asymmetric, identity-based, and group signature cryptography. To guarantee security and privacy, Al Shareeda et al. [4] investigated identity-based methods in VANETs. They also

discussed the entities involved and the VANET architectures. Syekh et al. [5] analyzed authentication schemes to protect vehicular networks from malicious nodes and fake messages. Mundhe et al. [6] analyzed cryptographic methods to meet the privacy, authentication, and other security requirements needed for VANETs. Jan et al. [7] analyzed and divided authentication and privacy methods into four main categories on the basis of their security procedures, security needs, strengths, weaknesses, countermeasures for attacks, and performance metrics. One of the most important aspects of the pseudonym life cycle is the pseudonym change process. Babaghayou et al. [8] provided an overview of the privacy problem and presented a review of recent and detailed state-of-the-art pseudonym change strategies and approaches. Farhood et al. [9] conducted study on VANETs and privacy concerns using pseudonym-changing strategies. Location privacy is another privacy concern. Khan et al. [10] examined location privacy in VANETs with security issues and conducted a critical study of attack and defense technologies. Table 1 presents a summary of various surveys on security and privacy in VANETs.

Despite the aforementioned literature reviews on VANET security, a full assessment of pseudonym life cycle implementation across several VANET system models is still lacking. Addressing this gap is crucial since each VANET entity has a unique purpose and interacts at different points of the pseudonym life cycle. A thorough and well-structured pseudonym life cycle, including issuance, usage, changing, resolution, and revocation, is required to achieve an optimal balance between privacy, security, and network efficiency. Without complete implementation, vehicles are vulnerable to linkability attacks, Sybil attacks, and illegal access, jeopardizing both anonymity and system security. Hence, building a well-integrated pseudonym a life cycle framework customized to various VANET architectures is critical for ensuring security, privacy-preserving communication and robust vehicular network management. This study aims to bridge this gap by analyzing the completeness of pseudonym life cycle implementations and identifying research challenges to facilitate future improvements in VANET security. We also analyze various types of VANET system models and link them to pseudonym life cycle implementation. Additionally, we identify privacy analyses conducted in relation to existing schemes to assess whether privacy requirements have been adequately addressed.

TABLE 1. Overview of various surveys on security and privacy in VANETs.

Survey	Year	Overview
Petit et al. [3]	2015	Provide a comprehensive survey of pseudonym approaches for vehicular networks
Al Shareeda et al. [4]	2021	Focus on identity-based methods in VANETs
Farhood et al. [9]	2024	Examine VANETs and privacy concerns using pseudonym-changing strategies
Khan et al. [10]	2021	Focus on location privacy in VANETs with security issues and conduct a critical study of attack and defense technologies
Mundhe et al. [6]	2021	Focus on cryptographic techniques proposed to achieve authentication, privacy, and other security features required in VANETs
Jan et al. [7]	2021	Provide a comprehensive survey of VANET security techniques by classifying schemes based on key characteristics
Babaghayou et al. [8]	2020	Introduce the privacy problem and provide a recent and detailed state of the art overview of the different suggested pseudonym change strategies and approaches
Syeikh et al. [5]	2019	Review authentication schemes that can protect vehicular networks from malicious nodes and fake messages
Propose a survey		We analyze and classify the VANET system model in relation to emerging technology support, identify the roles and mapping the incorporation of these technologies in the implementation of the pseudonym life cycle, analyze the implementation of the pseudonym life cycle stages, classify privacy analyses and identify research gaps and future research directions

The major contributions of this work include the following.

- We conducted a comprehensive survey of the state-of-the-art pseudonym life cycle implementations in VANETs, covering the issuance, usage, changing, resolution, and revocation stages. We classified the issuance stage on the basis of the entity responsible for generation, the input used, and the method for generating the pseudonym. For the changing stage, we classified the changing strategies in accordance with the ETSI TR 103415 standard. To our knowledge, there has not been a comprehensive survey or review of the pseudonym life cycle in VANETs within the past five years.
- We categorized VANET system models and their improvements with the support of emerging technologies. We mapped the incorporation of these technologies in the implementation of the pseudonym life cycle.
- We classified the privacy analyses conducted with respect to existing schemes. Specifically, we distinguished between informal analysis (e.g., compliance with privacy requirements and attack resistance discussions) and formal analysis techniques, such as the random oracle model (ROM), BAN logic, AVISPA, and ProVerif. Our classification highlights the extent of rigorous privacy assurance applied across different schemes and identifies underutilized yet promising formal methods.
- We identified gaps to address in future research and highlighted key factors for developing robust

privacy-preserving schemes in VANETs that deserve more attention.

This review is organized as follows. The method we adopt to conduct the review is described in Section II. Section III outlines our research questions, including trends in VANET system models, pseudonym life cycle implementation, and privacy analysis. Research challenges and future research directions are discussed in Section IV. The conclusions of this study are presented in Section V. The structure of our paper is illustrated in Figure 2. The figure also illustrates the logical flow of the review process, beginning with the introduction and the method (including planning, conducting, and reporting), followed by a discussion of system models, pseudonym life cycle trends, and privacy analysis. The process ends with an outline of research challenges and future directions and the conclusion.

II. METHODS

To structure this study, we adopted Kitchenham's guidelines [11] for conducting systematic reviews. This approach includes the following stages and activities: planning the review research questions, conducting the review search strategy, performing the study quality assessment, extracting data, synthesizing the data, and reporting the findings.

A. PLANNING THE REVIEW

The goal of this study is to methodically identify and analyze privacy schemes in the literature to address concerns regarding the implementation of the pseudonym life cycle in existing schemes. The following research questions (RQs) are addressed:

- RQ1 What are the trends in VANET system models?
- RQ2 What are the trends in pseudonym life cycle implementation in existing VANET schemes?
- RQ3 What types of privacy analyses are conducted to assess existing schemes?
- RQ4 What are the research challenges and potential gaps for future directions related to the complete implementation of the pseudonym life cycle in VANETs?

B. CONDUCTING THE REVIEW

In this section, we describe the systematic review process we conducted, covering the following stages: (1) identifying search strategies used to locate research papers in databases; (2) applying selection criteria to determine relevant research papers from the search results; (3) assessing the quality of the chosen research papers; and (4) extracting and synthesizing data to analyze and integrate the findings.

1) SEARCH STRATEGY

A comprehensive literature review was carried out to fully examine the state of research on privacy-preserving methods in VANETs. Three reputable bibliographic databases—Scopus, IEEE Explore, and Science Direct—were chosen as the main sources for data collection. These databases

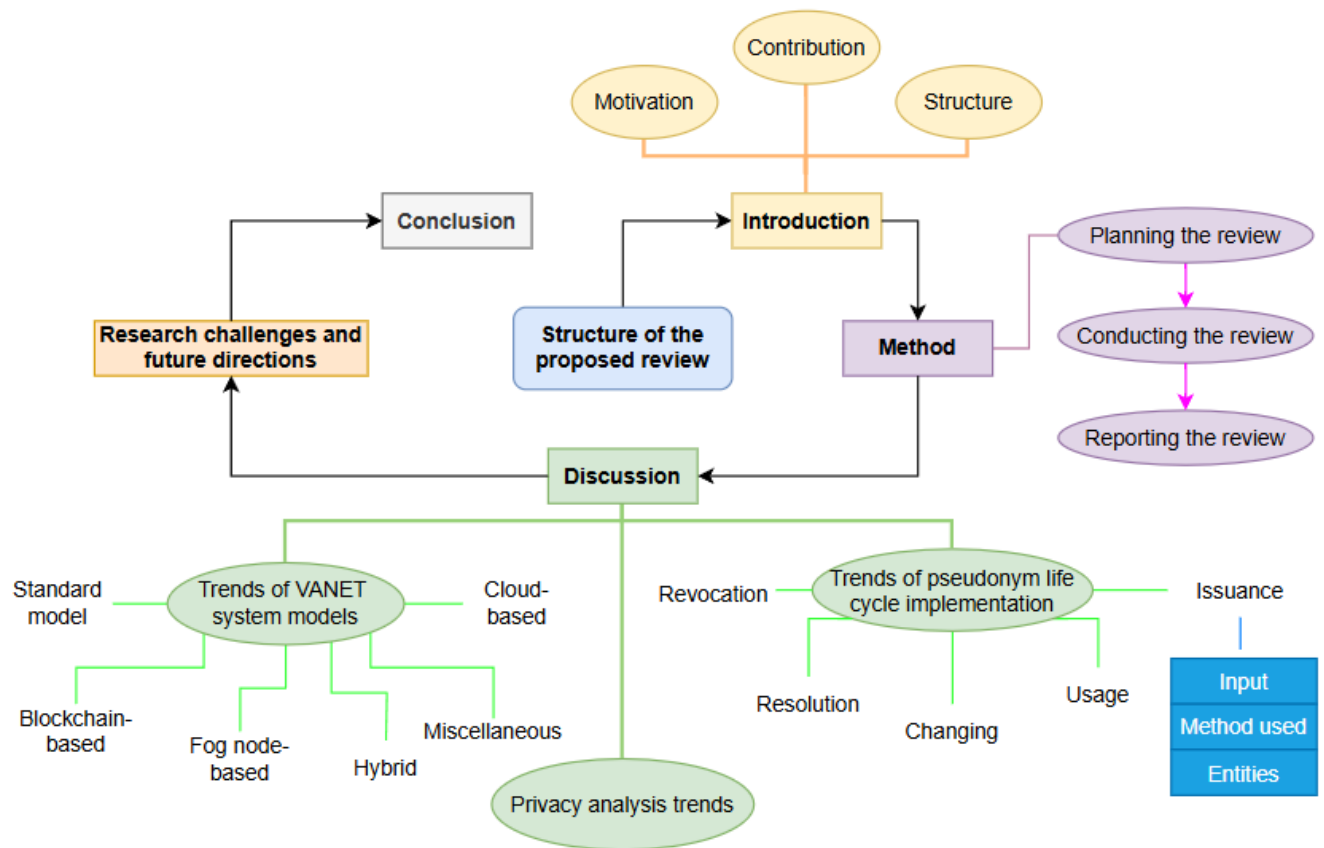


FIGURE 2. Structure of the proposed review.

TABLE 2. Search results.

Databases	Command Search	Date Filters	Search Date	Total
Scopus	Privacy-preserving in VANET	2019-2024	January, 5 2025	153
IEEE Explore	Privacy-preserving in VANET	2019-2024	January, 5 2025	171
Science Direct	Privacy-preserving in VANET	2019-2024	January, 5 2025	270

were chosen for their comprehensive coverage of high-quality peer-reviewed articles on VANETs, cybersecurity, and cryptography. The following keywords were used on the basis of the research questions: “privacy-preserving” and “VANET.” Using the Boolean operator “AND,” numerous search strings were carefully constructed to retrieve pertinent publications. For transparency and reproducibility, the final search results, together with the applied date filters and the entire set of search strings, are shown in Table 2.

2) SELECTION STRATEGY

Strict inclusion and exclusion criteria were established to ensure a thorough evaluation of the selected studies. The inclusion criteria prioritized high-quality empirical research. Studies were included if they were peer-reviewed journal articles published within the last six years in Science

Direct, IEEE Explore, or Scopus. The exclusion criteria, on the other hand, sought to eliminate studies that did not present novel research findings, such as book reviews, book chapters, position papers, analytical papers, comparative papers, technical reports, workshop reports, lecture notes, and conference papers. These types of publications were eliminated because they often lack rigorous peer review and may not reveal novel experimental or theoretical improvements. Similarly, conference papers were eliminated to emphasize high-impact journal articles, as conference papers may represent preliminary discoveries that are later extended in journal articles. Furthermore, studies whose full text was unavailable were excluded to ensure a comprehensive and thorough examination of methodologies, results, and conclusions. This approach is consistent with standard practices in systematic literature reviews, thus ensuring the reliability and reproducibility of our study.

3) QUALITY ASSESSMENT

A binary assessment criterion was established to systematically evaluate the included studies with respect to the development and use of pseudonyms. Two possible answers to a single evaluation question (AQ1) were formulated: “yes” and “no.” A “yes” response received a score of “1,” signifying the existence of a well-defined pseudonym

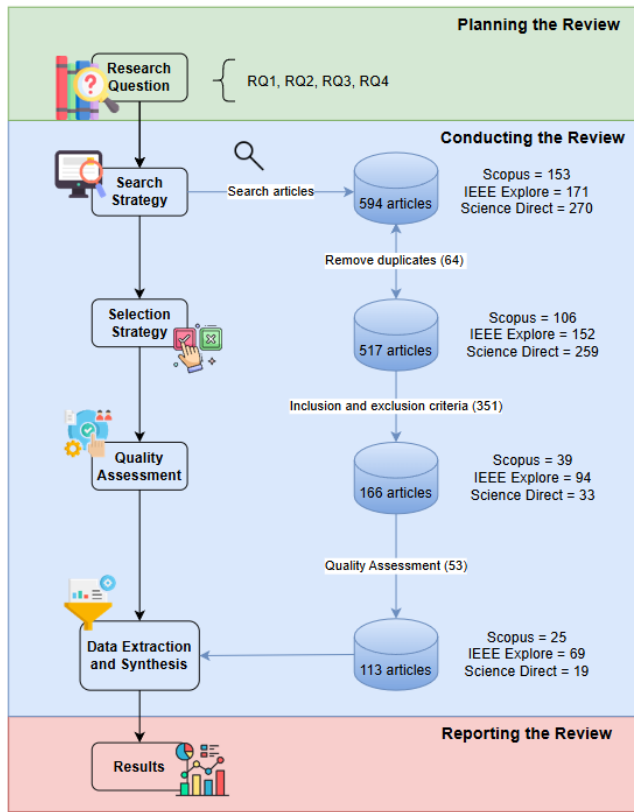


FIGURE 3. Systematic review process based on Kitchenham's methodology.

and the procedure used to create it to safeguard privacy. On the other hand, a “no” response was assigned a score of “0,” which meant that the study was not included in the subsequent analysis. Therefore, only studies that specifically discussed the creation and application of pseudonyms to improve privacy were considered for a thorough analysis.

In the first phase of our systematic literature review, we identified 594 papers that were potentially relevant to our research aims. To maintain the integrity of our review process, we first removed duplicate entries. Following careful screening, 64 duplicate papers were detected and eliminated, leaving a refined pool of 517 unique papers. Next, we used the inclusion and exclusion criteria to further reduce the number of papers. This thorough filtering method decreased the pool to 166 papers that were considered highly relevant to our research focus. Next, we conducted a quality assessment, resulting in a final list of 113 high-quality studies that served as primary research (PR), the foundation for our research. This methodical process ensured that our review is based on trustworthy and impactful studies, thus ensuring the credibility and validity of our conclusions. Figure 3 depicts the review process.

4) DATA EXTRACTION AND SYNTHESIS

In this study, we developed a structured data collection form as a standardized tool to guide the systematic extraction of relevant information from the selected research articles. The

TABLE 3. Extracted data.

Data	Property	Purpose	Section	Related-tables
D1	Research method	Overview	II-B	Table 2
D2	Distribution of PR	Overview	II-C	N/A
D3	VANET system model	RQ1	III-A	Table 4 - Table 8
D4	Pseudonym life cycle implementation	RQ2	III-B	Table 9 - Table 11
D5	Privacy analysis	RQ3	III-C	Table 12 - Table 13
D6	Research challenges and future research directions	RQ4	IV	Table 14

form encompassed key attributes such as VANET system models, the pseudonym life cycle, and privacy analysis. These attributes were carefully chosen to ensure a comprehensive and consistent approach to data extraction. Once the form was finalized, the selected articles were thoroughly reviewed, and the corresponding fields were populated accordingly. The extracted data were then organized into tables, which facilitated both comparison and thematic analysis. This structured workflow ensured consistency, traceability, and completeness throughout the data analysis process.

During the data extraction phase, we focused on key sections of each primary research study that aligned with the attributes listed in Table 3. Data for D1 and D2 were collected to provide an overview of the primary research studies, with an emphasis on the research methodologies and the distribution of privacy-related contributions. We recorded data for D3 to D6 to address our research questions in depth. To answer RQ1, we classified the VANET system models and further analyzed the advantages, challenges, and relevance of implementing the pseudonym life cycle within each model. For RQ2, we examined the pseudonym life cycle stages—issuance, usage, changing, resolution, and revocation—as implemented in the reviewed papers. Our analysis of the issuance stage identified the responsible entities, input data, and methods used for pseudonym generation. For the changing stage, we categorized the pseudonym change strategies based on ETSI TR 103 415 [12].

To address RQ3, we classified the privacy analyses conducted in the studies into two categories: formal and informal. This classification highlights the extent of rigorous privacy assurance applied across different schemes and identifies underutilized yet promising formal methods. Finally, in response to RQ4, we identified potential research gaps and outlined future research directions concerning the comprehensive implementation of the pseudonym life cycle in VANET environments.

C. REPORTING THE REVIEW

In this section, we present the yearly distribution of research papers and the data collection process used to address each research question.

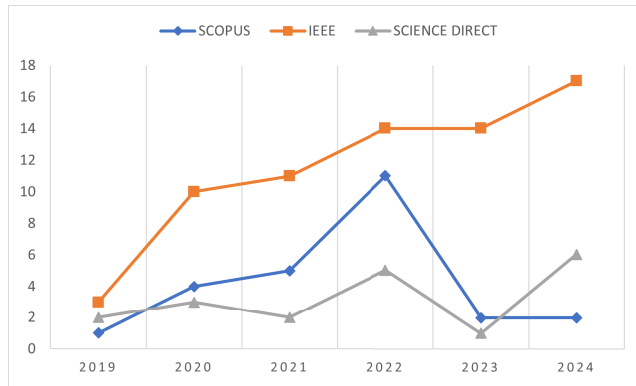


FIGURE 4. Distribution of journal studies across publication years.

Figure 4 displays the allocation of PR articles between 2019 and 2024. This graph contrasts the quantity of papers that are indexed in Science Direct, IEEE Explore, and Scopus. This analysis helped us determine which database is most widely used to publish research on privacy-preserving schemes in VANETs and to identify patterns in the increase or decrease in the annual number of publications. IEEE is the dominant source in all years, with the largest contribution in terms of the number of papers in all periods. This finding indicates that more privacy-preserving research on VANETs is published in IEEE-affiliated journals.

III. DISCUSSION

In this section, we answer RQ1 in Section III-A, which includes the trends of the VANET system model. We answer RQ2 in Section III-B, which includes the trends in pseudonym life cycle implementation, and in Section III-C, we respond to RQ3 about privacy analysis trends.

A. TRENDS IN THE VANET SYSTEM MODEL

VANET is a type of mobile ad hoc network (MANET) that allows vehicles to communicate with one another and with roadside infrastructure, thereby improving traffic safety, efficiency, and the provision of extra services such as navigation and entertainment. The VANET architecture consists of several components, communication modes and layers that collectively support safe and efficient communication. The basic components of the VANET system model or architecture include three entities: the vehicle/on-board unit (OBU), the roadside unit (RSU) and the trusted authority (TA). Figure 5 shows the VANET standard model.

OBUs are communication devices installed in vehicles. They enable vehicles to communicate with other vehicles, known as vehicle-to-vehicle (V2V) communication, and with the road infrastructure, known as vehicle-to-infrastructure (V2I) communication. OBUs are equipped with wireless communication interfaces, GPSs and other sensors that collect data such as the location, speed and environmental conditions of the vehicle. These units generate and receive information such as traffic warnings and road conditions.

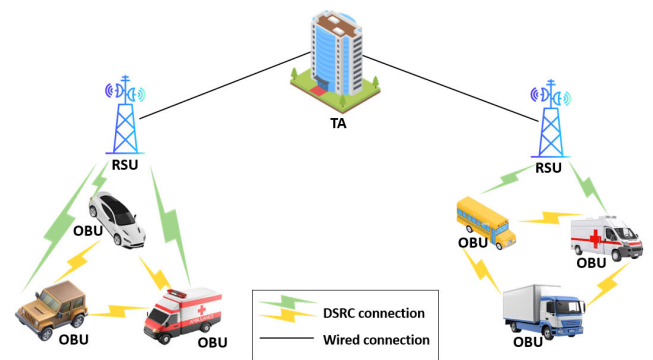


FIGURE 5. VANET standard model.

RSUs are fixed infrastructure units installed at key locations such as intersections or along highways to facilitate V2I communication. These units are connected to back-end servers or traffic management centers via wired or wireless networks, allowing them to relay traffic data, manage traffic signals or send messages to vehicles (e.g., regarding road hazards or congestion). The TA is a central authority responsible for the management and security of communications within VANETs. It ensures that only authorized entities participate in the network by issuing security credentials such as digital certificates and pseudonyms to vehicles. As VANET architectures continue to evolve, this basic model can be integrated with emerging technologies to improve efficiency and security. We identified six VANET system models in Table 4.

1) STANDARD MODEL

Table 3 highlights the prevalence of the standard model over other architectures, with approximately 71% (80 of 113) of the reviewed research papers employing this approach. The standard model continues to be the preferred choice in VANET systems because of its relatively simple and efficient architecture, which facilitates practical implementation. One of the key advantages of this model lies in its centralized security framework, where the TA serves as the guarantor of privacy and credential management. Additionally, the TA plays a pivotal role in supporting law enforcement efforts to trace and identify malicious vehicles. However, this centralized security model faces a critical challenge in the form of a single point of failure (SPoF), such that the unavailability of the TA affects all entities within the VANET. Furthermore, given that the TA has knowledge of the identities of all network entities, it becomes a prime target for privacy breaches. A compromised TA could therefore result in widespread privacy violations, raising significant security concerns.

Several PR studies have improved the standard model by dividing the TA's authority to solve the SPoF problem. The TA should be split into two subauthorities, the trace authority (TRA) and the key generation center (KGC), according to PR30 [33], PR53 [51], PR37 [37], PR67 [60], and PR93 [80].

TABLE 4. VANET system model and associated primary research (PR) sources.

Models	Source	Total	Freq.
Standard	PR01 [13], PR03 [14], PR04 [15], PR05 [16], PR06 [17], PR08 [18], PR09 [19], PR10 [20], PR13 [21], PR14 [22], PR15 [23], PR16 [24], PR17 [25], PR19 [26], PR20 [27], PR21 [28], PR23 [29], PR26 [30], PR28 [31], PR29 [32], PR30 [33], PR31 [34], PR32 [35], PR33 [36], PR37 [37], PR38 [38], PR39 [39], PR41 [40], PR42 [41], PR44 [42], PR45 [43], PR46 [44], PR47 [45], PR48 [46], PR49 [47], PR50 [48], PR51 [49], PR52 [50], PR53 [51], PR55 [52], PR56 [53], PR57 [54], PR58 [55], PR62 [56], PR63 [57], PR64 [58], PR65 [59], PR67 [60], PR68 [61], PR69 [62], PR70 [63], PR71 [64], PR72 [65], PR73 [66], PR74 [67], PR75 [68], PR76 [69], PR77 [70], PR78 [71], PR79 [72], PR81 [73], PR82 [74], PR83 [75], PR84 [76], PR86 [77], PR88 [78], PR90 [79], PR93 [80], PR94 [81], PR95 [82], PR96 [83], PR98 [84], PR100 [85], PR101 [85], PR103 [86], PR105 [87], PR107 [88], PR108 [89], PR110 [90], PR112 [91], PR113 [92]	80	80/113
Blockchain-based	PR22 [93], PR24 [94], PR27 [95], PR35 [96], PR36 [97], PR40 [98], PR66 [99], PR80 [100], PR87 [101], PR91 [102], PR99 [103], PR102 [104], PR104 [105], PR106 [106], PR111 [107]	15	15/113
Cloud-based	PR02 [108], PR12 [109], PR54 [110], PR59 [111], PR85 [112], PR109 [113]	6	6/113
Fog-based	PR25 [114], PR61 [115], PR89 [116], PR97 [117], PR98 [118]	5	5/113
Hybrid Model	PR07 [119], PR34 [120], PR43 [121], PR60 [122], PR92 [123]	5	5/113
Miscellaneous	PR11 [124], PR18 [125]	2	2/113

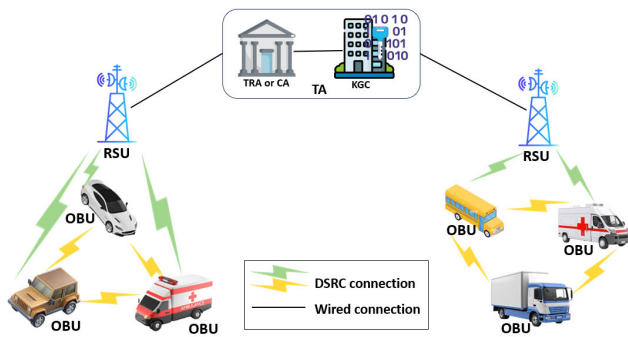


FIGURE 6. VANET standard model with subauthorities: the TA is divided into a trace authority (TRA or CA) and a key generation center (KGC).

While the KGC manages the generation of cryptographic keys and security parameters needed by the system, the TRA is responsible for generating pseudononyms and tracking rogue vehicles. However, both the TRA and the KGC still operate under the overarching authority of the TA. The system model that allocates the TA's authority is shown in Figure 6. This separation enhances security by preventing a single entity from having full control over privacy-sensitive information, thus making it harder for attackers to compromise the system.

Rather than distributing the authority of the TA, some PR studies incorporate separate entities that inherently address the SPoF issue in relation to the TA. While this approach differs from direct TA distribution, it effectively delegates TA functions to independent entities. These entities include the KGC in PR31 [34], PR51 [49], PR71 [64], PR107 [88], and PR108 [89]; the central authority (CtA) in PR88 [78]; the regional trusted authority (RTA) in PR52 [50]; the regional traffic management center (RTMC) in PR75 [68]; the certification authority (CA) in PR100 [84]; and the root trusted authority (RoTA) in PR74 [67]. The

KGC serves the same function as previously described, generating cryptographic keys and security parameters. The CtA is responsible for managing all certificate requests from vehicles, whereas the RTA functions as a local TA for generating initial pseudonyms. The CA is responsible for initially registering vehicles and providing temporary pseudononyms. The RTMC is employed to generate temporary secret keys, whereas the RoTA operates as the top-layer authority, authorizing and issuing certificates while also generating public and private keys for the TA.

In addition to introducing new entities to mitigate the SPoF issue in relation to the TA, several PR studies have incorporated additional entities aimed at enhancing the effectiveness and efficiency of VANETs. These entities include the service provider (SP) in PR63 [57]; the application server (AS) in PR32 [35]; PR33 [36]; PR55 [52]; PR70 [63]; PR113 [92]; the cooperative vehicle (CV) in PR29 [32]; the local inspector (LI) in PR100 [84]; and the traffic management center (TMC) in PR41 [40] and PR112 [91]. The SP is responsible for connecting all RSUs and storing feedback from vehicles. The AS receives traffic information from RSUs, predicts road conditions, and supports RSUs in providing security messages and safety applications. The CV assists RSUs in verifying messages to reduce latency. The LI serves as a locally placed unit responsible for V2I authenticated key sharing and providing traffic data to local vehicles. Finally, the TMC performs message verification and analysis to extract valuable traffic information. Figure 7 shows the entity development of the standard VANET model.

In addition to the addition of new entities to the VANET, several PR studies have enhanced the capabilities of the OBU by incorporating technologies such as biometrics, tamper-proofing, and physical unclonability (PUF). In PR08 [18], biometric technology was used to support multifactor vehicle authentication, and a tamper-proof device (TPD) was used to store sensitive information, including secret keys, group keys,

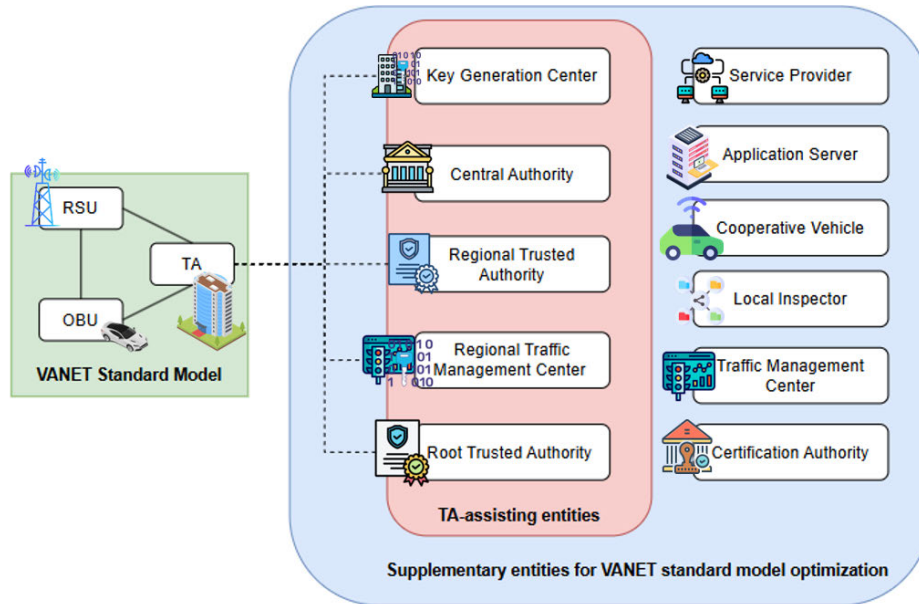


FIGURE 7. Incorporation of entities within the VANET standard model.

and biological passwords. The stored group key serves as one of the inputs for pseudonym generation, a process handled by the OBU. In line with the role of the TPD as a medium for storing sensitive data, PR28 [31] also used the TPD to store a vehicle's public key, which acts as an input for pseudonym generation by the OBU. Moreover, the function of the PUF in PR05 [16] and PR90 [79] is to generate random outputs to support the vehicle authentication process.

In the pseudonym lifecycle, the addition of entities in the standard VANET model enhances the implementation of the pseudonym lifecycle. These additional entities help distribute the responsibilities associated with different stages of the lifecycle, including issuance, usage, change, resolution, and revocation. The TRA, as described in PR37 [37], PR53 [51], PR67 [60], PR100 [84] and PR112 [91], plays a crucial role in pseudonym issuance and resolution. It is responsible for issuing pseudonimities to vehicles while ensuring that these pseudonyms can be efficiently resolved when necessary, such as during security investigations or law enforcement scenarios. By handling both pseudonym issuance and traceability, the TRA enhances the balance between privacy protection and accountability in VANETs, mitigating the risk of unauthorized tracking while still enabling authorities to identify malicious vehicles when needed. Other entities, such as the KGC in PR69 [62], RTA in PR52 [50] and TPD, as proposed in PR08 [18] and PR28 [31], also contribute to pseudonym issuance. By integrating these entities, the pseudonym lifecycle in VANETs becomes more efficient, secure, and resilient against potential vulnerabilities.

2) BLOCKCHAIN-BASED MODEL

The adoption of blockchain-based architectures in 13% (15 of 113) of PR studies utilize blockchain-based architectures

highlights the growing interest of researchers in more decentralized and scalable models. This trend reflects a shift toward architectures that aim to address the inherent limitations of VANETs, particularly centralization. By embracing decentralization, reliance on the TA as a single authority can be minimized, thereby enhancing system transparency. However, despite these advantages, blockchain implementation poses challenges related to computational overhead during the consensus process, which may not be suitable for resource-constrained devices such as OBUs. Additionally, the latency introduced by consensus mechanisms may limit communication efficiency.

In addition to decentralization, blockchain technology offers several key advantages, including trust management (as demonstrated in PR22 [93] and PR35 [96]), on-chain certificate management (PR27 [95]), secure pseudonym distribution and batch pseudonym revocation (PR40 [98]), and revocation transparency (PR27 [95], PR35 [96], and PR36 [97]). Smart contracts have been used to improve privacy in PR24 [94], PR27 [95], PR35 [96], and PR80 [100]; assist with forensic investigations in PR106 [106]; and enable dynamic revocation in PR111 [107]. Furthermore, we identify the critical role of blockchain technology in enhancing privacy and security within VANETs. The key roles (KRs) of the blockchain in VANET security and privacy include the following:

- Decentralization (KR1): Blockchain technology enables the creation of a decentralized authentication system, eliminating the need for a centralized third party, which frequently presents the risk of an SPoF and raises privacy concerns.
- Immutability (KR2): The immutability of blockchain technology means that once data are recorded, they

cannot be changed or tampered with, resulting in a high level of integrity in VANET connections. This feature is essential for ensuring data authenticity and privacy since it prohibits malicious modifications or unauthorized access to critical information.

- Trust management (KR3): A blockchain-based trust management model is critical for guaranteeing that messages transmitted within VANETs are synced and trustworthy. This paradigm improves the integrity and authenticity of vehicle communications by exploiting blockchain technology's decentralized, tamper-resistant, and transparent characteristics. Blockchain-based consensus methods ensure that only confirmed and authenticated messages are distributed across the network, reducing the risks associated with false information propagation, Sybil attacks, and message manipulation.
- Reputation synchronization (KR4): A blockchain allows a safe reputation value update mechanism that employs a distributed consensus process to ensure synchronized reputation evaluations across the network. This decentralized technique improves trustworthiness in trust management systems by eliminating manipulation and guaranteeing that reputation scores are accurate, transparent, and tamper-proof.
- Conditional privacy and traceability (KR5): A blockchain can help to build a privacy-preserving authentication system, ensuring that vehicles are verified without revealing their true identities to unauthorized entities, thus allowing authorities to track and identify harmful cars as needed. This balance protects privacy while ensuring accountability and security within the network.
- Anonymity and Pseudo-ID (KR6): A blockchain can be used as a publicly verifiable record for pseudonimities that have been issued but not revoked, resulting in a safe and transparent identity management process. This approach uses blockchain's immutability and decentralization to enable rapid verification of pseudonimities without exposing the sender's true identity, hence improving privacy.
- Smart contract for privacy (KR7): Using smart contracts on a blockchain allows for automatic and safe transactions. These smart contracts are programmed to automatically enforce privacy policies and restrictions, ensuring that data are released only in established, trusted circumstances.
- Public key information management (KR8): A blockchain uses smart contracts to ensure secure administration of anonymous public keys for automobiles and RSUs. These contracts enable the system to store, retrieve, and revoke public keys depending on access rights, which guarantees that only authorized entities can edit key data.
- On-chain certification management (KR9): A blockchain improves on-chain certificate management

for VANET security and privacy by offering a decentralized, immutable, and transparent framework for certificate issuance, verification, and revocation. Immutable blockchain records ensure tamper-proof certificate storage, whereas on-chain revocation lists allow for real-time changes, thus preventing the usage of compromised certificates.

- Batch pseudonym revocation (KR10): A blockchain records the pseudonym distribution process, allowing for accurate tracing and revocation of pseudonyms within the same batch. This capacity is critical for maintaining privacy since it enables effective management of pseudonyms without disclosing individual identities. Authorities may trace and revoke pseudonyms securely using the blockchain's transparent and immutable ledger, thus ensuring privacy while maintaining system integrity and accountability.
- Secure pseudonym distribution (KR11): A blockchain's pseudonym distribution records include necessary information, such as the sender, receiver, and timestamp, while ensuring that no private or sensitive information is published. The use of a secure hash function with strong collision resistance is critical in maintaining the confidentiality of the original data.
- Revocation transparency (KR12): The use of blockchain technology allows for fast revocation and transparency of pseudonimities. This approach ensures that even when a pseudonimity is revoked, the process is transparent and protects the privacy of the vehicle concerned. The unchangeable ledger of a blockchain ensures that revocations are reliably recorded, which provides a clear audit trail without disclosing sensitive vehicle information.
- Smart contracts for forensics (KR13): Smart contracts facilitate electronic forensics and vehicle tracing in accident scenarios by leveraging predefined conditions stored on a blockchain. This process ensures a systematic, transparent, and tamper-proof approach to incident investigations, enhancing accountability and trust in VANET security.
- Smart contracts for dynamic revocation (KR14): This role allows for efficient and scalable management of vehicle identities, ensuring that unauthorized vehicles can be tracked and revoked without relying on a central authority.

Next, we conduct a thorough analysis of fourteen KRs identified in research papers that integrate blockchain technology into VANETs. This analysis provides a detailed perspective on how blockchain technology strengthens security and privacy, thus offering valuable insights into the effectiveness of blockchain-based solutions in preserving security and privacy in VANETs. Table 5 outlines the key role of blockchain technology and the related research sources.

Next, we map the relationship between key blockchain roles and the pseudonym life cycle in VANETs. This step is

TABLE 5. Blockchain-enabled security and privacy enhancements in VANETs: Key roles and related primary research (PR) sources.

Sources	KR1	KR2	KR3	KR4	KR5	KR6	KR7	KR8	KR9	KR10	KR11	KR12	KR13	KR14
PR22 [93]	A	A	A	A	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
PR24 [94]	A	A	N/A	N/A	A	A	A	A	N/A	N/A	N/A	N/A	N/A	N/A
PR27 [95]	A	A	N/A	N/A	A	A	A	N/A	A	N/A	N/A	A	N/A	N/A
PR35 [96]	A	A	N/A	N/A	A	A	A	A	N/A	N/A	N/A	A	N/A	N/A
PR36 [97]	A	A	A	N/A	A	A	N/A	N/A	N/A	N/A	N/A	A	N/A	N/A
PR40 [98]	A	A	N/A	N/A	A	A	N/A	N/A	N/A	A	A	N/A	N/A	N/A
PR66 [99]	A	A	N/A	N/A	A	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
PR80 [100]	A	A	N/A	N/A	A	A	A	A	N/A	N/A	A	A	N/A	N/A
PR87 [101]	A	A	N/A	N/A	A	A	N/A	N/A	N/A	N/A	N/A	A	N/A	N/A
PR91 [102]	A	A	N/A	N/A	A	A	A	N/A	N/A	N/A	N/A	A	N/A	N/A
PR99 [103]	A	A	N/A	N/A	A	A	N/A	N/A	N/A	N/A	N/A	A	N/A	N/A
PR102 [104]	A	A	N/A	N/A	A	A	N/A	A	N/A	N/A	N/A	A	N/A	N/A
PR104 [105]	A	A	N/A	N/A	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
PR106 [106]	A	A	N/A	N/A	A	A	N/A	N/A	N/A	N/A	N/A	N/A	A	N/A
PR111 [107]	A	A	N/A	N/A	A	A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	A

A: Available, N/A: Not Available

performed not only to provide not only a structured understanding of blockchain integration in VANETs but also to identify solutions to challenges in pseudonym management, to guide the design of efficient architectures, to highlight the advantages of blockchain over conventional mechanisms, and to open new research directions in the development of blockchain-based security and privacy systems. Our analysis indicate that the blockchain has a key role in supporting the implementation of the pseudonym life cycle in the issuance, usage, resolution, and revocation stages. However, no suitable role of the blockchain has been identified for the changing stage, as certain inherent characteristics of the blockchain, namely, immutability, high latency, linkability risks, and limitations in off-chain mechanisms, are not well suited for enabling the rapid pseudonym changes required in VANETs. The mapping is presented in Figure 8.

3) CLOUD-BASED MODEL

Cloud computing has been used in six PR studies (5.3%), providing centralized yet scalable computational resources for controlling security and privacy in VANETs. Some significant benefits of using cloud computing in VANETs include centralized data processing, which enables efficient identity management and pseudonym-based security measures, and data storage scalability, which allows for flexible and dynamic handling of vehicle data.

However, integrating cloud computing into VANETs has led to several challenges. One major issue is high latency, as communication between vehicles and cloud servers may introduce delays that affect real-time system performance. Another concern is the SPoF, which occurs when a cloud infrastructure vulnerability causes service outages across the entire network. Additionally, privacy concerns exist, as sensitive vehicular data stored in the cloud are susceptible to unauthorized access and exploitation.

We investigate the key functions of cloud servers in these studies to learn about how cloud computing has been used to improve security and privacy in VANETs. Understanding these fundamental roles allows us to analyze

the merits and limitations of cloud-based approaches to ensure efficient, scalable, and privacy-preserving VANET systems. Table 6 presents a summary of the key functions of cloud servers in VANET security and privacy, along with the primary research papers that have implemented these functions. By mapping each function to the corresponding primary research studies, we aim to highlight the practical applications and effectiveness of cloud-based solutions in addressing security and privacy challenges in VANETs. The key functions (KFs) of the cloud server are as follows:

- Data processing (KF1): The cloud server processes the data collected by RSUs and makes them available to other cars. This function allows vehicles to access essential traffic and road condition information as needed, thereby increasing the overall efficiency of vehicle communication.
- Data storage (KF2): The cloud server stores safety messages collected from various RSUs. This function supports detailed traffic analysis and assists in changing traffic tactics on the basis of the data acquired.
- Security and privacy preservation (KF3): The cloud server is intended to operate in a semitrusted environment, meaning that it must protect user data from potential breaches. This function ensures that neither the vehicular users' private location information nor the navigation results are revealed, even in the presence of collaborating entities such as illegal users.
- Computational efficiency (KF4): The cloud server reduces the computational burden on vehicles and RSUs by performing complex calculations and data aggregation operations. Considering the restricted processing capabilities of standard VANET systems, this function is especially essential.
- Key generation and distribution (KF5): The cloud server facilitates secure communication and authentication inside the vehicular network by generating public and master keys. Once a secret key is generated, it is encrypted with an attribute key before the ciphertext is sent to the appropriate vehicle. This function ensures

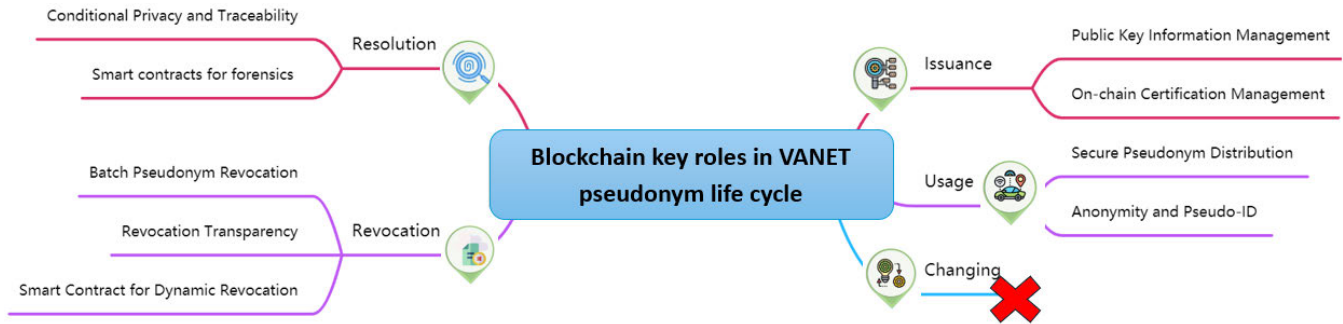


FIGURE 8. Key blockchain roles in the VANET pseudonym life cycle.

- that only the intended vehicle can decode and access the secret key, which improves security.
- Vehicle registration (KF6): The cloud server tracks all vehicle IDs, chassis numbers, and RSU IDs. This registration process is critical for retaining a database of vehicles and their related keys, which enables secure communication.
 - Facilitating secure communication (KF7): The cloud server establishes secure connections with RSUs, protecting transferred data from potential eavesdropping or tampering. Secure communication is essential for maintaining the integrity of the automotive network.
 - Security analysis (KF8): The cloud server also contributes to security analysis by detecting hostile nodes attempting to access the network and manage fake messages. This function ensures that the nodes are trustworthy, the system's integrity is preserved, and any fraudulent messages are immediately addressed.
 - Reduced communication costs (KF9): When a hybrid aggregation-verification technique is used, the cloud server considerably reduces the communication and storage costs associated with delivering numerous signatures to the cloud. The approach enables a single reaggregated signature, which is more efficient than delivering many large signatures.
 - Support for hierarchical architecture (KF10): The cloud server is a component of a hierarchical aggregation-verification system that includes cars and RSUs. This structure enables the efficient handling and verification of safety messages across various levels of the network.
 - Reputation calculation (KF11): The cloud server measures vehicle reputation on the basis of the aggregated feedback it receives. It may collect the entire number of vehicles and their relevant parameters from the feedback, which is critical for evaluating vehicle reliability and safety.
 - Service provision (KF12): The cloud server adds value to vehicles by offering a variety of applications that improve transportation efficiency and safety. These services are refined to satisfy the individual demands

TABLE 6. Cloud server-enabled security and privacy enhancements in VANETs: Key functions and related primary research (PR) sources.

Key Functions	PR2 [108]	PR12 [109]	PR54 [110]	PR59 [111]	PR85 [112]	PR109 [113]
KF1	A	N/A	A	A	A	A
KF2	A	N/A	A	A	A	A
KF3	A	N/A	N/A	N/A	N/A	N/A
KF4	A	N/A	N/A	A	A	N/A
KF5	N/A	A	N/A	N/A	N/A	N/A
KF6	N/A	A	N/A	N/A	N/A	N/A
KF7	N/A	A	N/A	A	A	N/A
KF8	N/A	A	A	N/A	A	N/A
KF9	N/A	N/A	A	N/A	A	N/A
KF10	N/A	N/A	A	N/A	N/A	N/A
KF11	N/A	N/A	N/A	A	N/A	N/A
KF12	N/A	N/A	N/A	N/A	N/A	A
KF13	N/A	N/A	N/A	N/A	N/A	A
KF14	A	A	N/A	N/A	N/A	N/A

A: Available, N/A: Not Available

- of drivers, thus increasing the overall usefulness of the VANET system.
- Data swapping (KF13): The cloud server also permits data swapping between vehicles and RSUs. This function is essential for maintaining real-time communication and ensuring that vehicles have access to the most recent traffic information, which can help with navigation and decision-making.
 - Support cryptographic operation (KF14): The cloud server utilizes various cryptographic techniques or collaborates with a cryptographic service provider. This cryptographic capability is vital for protecting sensitive data and maintaining privacy within the vehicle-based cloud computing environment.

Next, we present a detailed map of key cloud computing functions and the pseudonym life cycle in VANETs. This mapping not only provides a structured understanding of cloud computing implementation but also contributes to identifying potential solutions to pseudonym management challenges, informs the design of efficient architectures, emphasizes the benefits of cloud-based mechanisms over traditional approaches, and opens new research directions in the development of cloud-based security and privacy

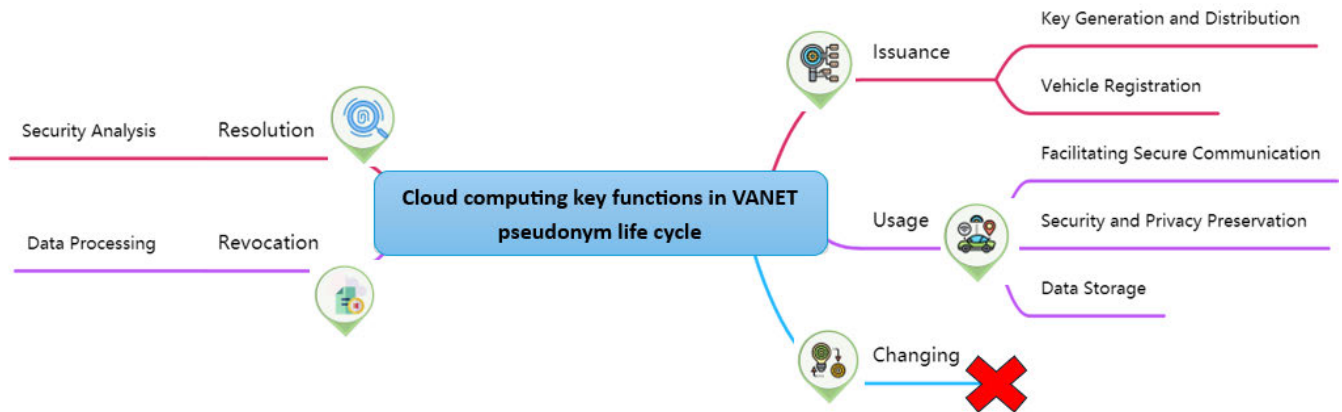


FIGURE 9. Cloud computing key functions in the VANET pseudonym life cycle.

systems for VANETs. Our investigation suggests that key cloud computing jobs involve primarily implementing the pseudonym life cycle during the issuance, usage, resolution, and revocation stages. However, no appropriate cloud server role has been discovered for the pseudonym-changing stage. This finding is supported by the lack of a shifting strategy mechanism in all six cloud-based research publications examined; this could be linked to communication latency, which restricts real-time pseudonym switching, the reliance on unstable network connectivity, and the hazards associated with centralization, thus increasing security vulnerabilities. The mapping is shown in Figure 9.

4) FOG-BASED MODEL

The goal of a fog-based approach is to improve real-time data processing by reducing latency and facilitating processing closer to the network edge. By using fog nodes close to vehicles to distribute computational duties, real-time decision-making improves, and communication delays decrease. Applications that rely on latency, such as real-time traffic management and collision avoidance, greatly benefit from this approach. However, problems still exist. Fog computing addresses issues with data consistency and synchronization across dispersed fog nodes. The need to safeguard data handled at the edge from intrusion and manipulation also presents security concerns.

To understand how fog nodes contribute to increasing security and privacy in VANETs, we investigated the primary tasks of fog nodes in 4.4% (5 of 113) of the PR studies that implemented fog nodes in VANETs. Understanding the primary functions of fog nodes allows for an evaluation of the benefits and drawbacks of fog node strategies to guarantee effective, scalable, and privacy-preserving VANET systems. The primary task fog nodes in VANET security and privacy are summarized in Table 7, which includes research publications that have implemented them. We hope that connecting each task to relevant research projects helps to demonstrate the usefulness and practical applicability of fog node-based solutions in resolving security and privacy

issues in VANETs. The fog node's primary tasks (PTs) are as follows:

- Authentication (PT1): Authentication ensures that only legitimate vehicle nodes are allowed to communicate within the network and/or that messages are new and legitimate. Some implementations also enable mutual authentication between vehicles and fog nodes.
- Facilitating communication (PT2): The fog node enables communication between the TA, OBU, and other entities within VANETs.
- Key or security parameter management support (PT3): The fog node assists the system in generating, verifying, and distributing keys or other security parameters within VANETs.
- Mitigating attack risk (PT4): By obfuscating information and coordinating secure communications, the fog node reduces the risk of attackers inferring sensitive information about the vehicle's purpose or the driver's identity.
- Reducing dependency on TAs (PT5): The fog node's role in processing and managing requests allows for reduced dependency on TAs, enhancing scalability and increasing system resilience against potential attacks.
- Minimizing latency (PT6): Fog nodes are strategically deployed at the edge of the vehicular network, which helps reduce latency. This placement is essential for real-time applications in VANETs, where timely data processing is critical for safety and efficiency.
- Enhanced computational capabilities (PT7): Fog nodes exhibit greater computational power than traditional RSUs do. This increased power allows fog nodes to handle more complex security tasks, such as node and message authentication, effectively improving the overall system throughput.
- Message integrity (PT8): Fog nodes help ensure the integrity of messages transmitted within the network, which is vital for preventing unauthorized access and ensuring that messages are not tampered with during transmission.

TABLE 7. Fog node-enabled security and privacy enhancements in VANETs: Primary tasks and related primary research (PR) sources.

Primary Tasks	PR25 [114]	PR61 [115]	PR89 [116]	PR97 [117]	PR98 [118]
PT1	A	N/A	N/A	A	A
PT2	A	N/A	A	A	N/A
PT3	A	A	N/A	N/A	N/A
PT4	N/A	N/A	A	N/A	A
PT5	A	A	A	A	A
PT6	N/A	A	N/A	A	N/A
PT7	N/A	A	N/A	N/A	A
PT8	N/A	N/A	N/A	N/A	N/A
PT9	N/A	A	N/A	N/A	A
PT10	N/A	A	N/A	N/A	N/A

A: Available, N/A: Not Available

- Facilitating conditional privacy (PT9): Fog nodes help maintain the anonymity of vehicles by allowing them to communicate without revealing their real identities, which is essential to prevent attackers from tracing vehicle information through communication channels.
- Revocation of malicious vehicles (PT10): Fog nodes enable quick revocation of malicious vehicles by allowing TAs to release only two hash seeds to invalidate all unexpired pseudonyms of a problematic vehicle. This mechanism ensures that the network can respond quickly to security threats.

We relate the specified primary tasks to the implementation of the pseudonym life cycle in fog-based VANETs. This mapping is critical for assessing security and privacy effectiveness, identifying implementation gaps, and enhancing system design. Additionally, this mapping aids in determining the extent to which fog nodes enable critical steps such as issuance, usage, resolution, revocation, and alteration, resulting in improved pseudonym management; furthermore, it eliminates reliance on the TA, thus increasing system efficiency and lowering communication latency. Precisely defining the function of fog nodes in pseudonym life cycle management supports the development of more distributed, scalable, and efficient VANET architectures. We found that fog nodes are essential to certain pseudonym life cycle stages, namely, issuance, usage, resolution, and revocation. However, their function is not apparent during the changing stage. Although pseudonym changes are implemented in PR 43 [121], this function is carried out by the vehicle itself rather than by the fog node. The mapping of the primary tasks of the fog node in the pseudonym life cycle is presented in Figure 10.

5) HYBRID MODEL

Another model identified in this review is the hybrid model, which combines the strengths of cloud, fog, and blockchain technologies to enhance privacy preservation. This hybrid model leverages the scalability of cloud computing, the low latency of fog computing, and the decentralized trust of the blockchain, potentially offering a more robust solution for VANET security. Within this model, blockchain technology

secures data transactions, fog technology supports fast and localized processing, and cloud technology manages large-scale data storage and complex analytics. In PR07 [119], the strengths of cloud, fog, and blockchain technologies are integrated, enhancing both security and privacy by providing decentralized trust management, immutable records, real-time processing, and efficient resource management. In PR34 [120], fog and blockchain technologies are integrated. The blockchain provides a secure and reliable framework for managing vehicle identities and transactions, whereas fog computing enhances the efficiency and privacy of traffic route management. Together, these technologies facilitate the creation of a robust system that addresses the security vulnerabilities and high computational costs associated with traditional vehicular networks. In PR43 [121] and PR60 [122], fog and cloud technologies are integrated. In PR43 [121], fog nodes focus on real-time authentication and privacy preservation, whereas cloud servers provide centralized management and analysis capabilities, together forming a comprehensive security and privacy framework for VANETs. In PR60 [122], fog nodes act as the first line of processing and aggregation for real-time data from vehicles, whereas cloud servers provide the necessary infrastructure for storage and complex data processing, ensuring the overall efficiency and effectiveness of the safety warning system in fog-cloud-based VANETs. In PR92 [123], cloud and blockchain technologies are integrated. The cloud server is essential for data processing and storage, whereas blockchain technology provides a robust framework for ensuring data integrity and supervising the cloud server's actions. Together, they facilitate the creation of a more secure and efficient system for monitoring road conditions in VANETs.

In Table 8, we present an overview of the roles and functions of cloud servers, fog nodes, and blockchain technologies to improve the understanding of their contributions to VANET architectures.

However, integration challenges arise. The integration of these three distinct technologies increases system complexity, and ensuring smooth coordination between the cloud, fog, and blockchain components requires advanced control mechanisms, which could lead to higher costs and difficulties in managing interoperability.

6) MISCELLANEOUS

In addition to hybrid models, we identified research papers focusing on Internet of Things (IoT)-based [125] and software-defined vehicular network (SDVN)-based [124] models, albeit with low frequency (1 of 113 PR papers). IoT integration allows vehicles to interact with a wide range of devices and systems, thus enabling richer data collection and supporting advanced applications such as smart traffic management and accident prevention. The challenge for IoT-based models lies in the heterogeneous nature of IoT devices, which introduces security vulnerabilities, as each device may have different trustworthiness and capabilities. Protecting privacy and ensuring secure communication

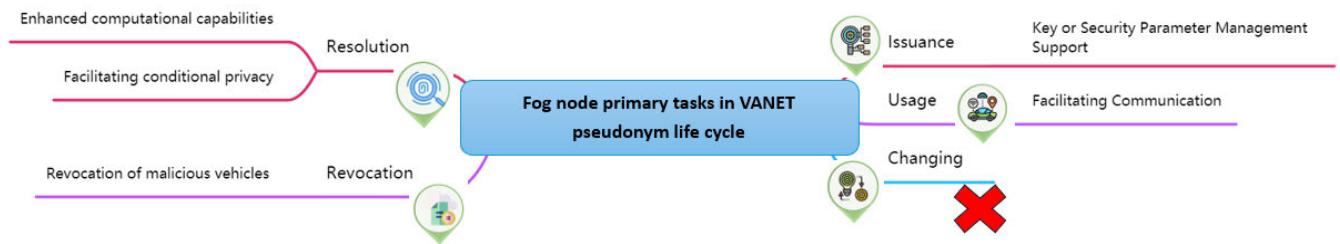


FIGURE 10. Primary tasks of fog nodes in the VANET pseudonym life cycle.

TABLE 8. Blockchain-, cloud server- and fog node-enabled security and privacy enhancements in VANETs: Roles/functions/tasks and related primary research (PR) sources.

Roles/functions/tasks	Sources
Decentralization (KR1)	PR07 [119], PR34 [120], PR92 [123]
Immutability (KR2)	PR07 [119], PR34 [120], PR92 [123]
Trust management (KR3)	PR07 [119]
Public key information management (KR8)	PR34 [120]
Revocation transparency (KR12)	PR07 [119]
Data processing (KF1)	PR07 [119], PR60 [122], PR92 [123]
Data storage (KF2)	PR07 [119], PR43 [121], PR60 [122], PR92 [123]
Security and privacy preservation (KF3)	PR43 [121]
Computational efficiency (KF4)	PR60 [122]
Authentication (PT1)	PR43 [121]
Minimizing latency (PT6)	PR07 [119], PR34 [120], PR60 [122]
Enhanced computational capabilities (PT7)	PR07 [119], PR34 [120]
Message integrity (PT8)	PR43 [121], PR60 [122]
Facilitating conditional privacy (PT9)	PR43 [121]

among numerous devices pose significant challenges. Furthermore, SDVNs apply software-defined networking (SDN) principles to VANETs, thereby enhancing network flexibility and management. This model offers more dynamic control over the network, allowing for rapid implementation of security policies and efficient resource allocation. The programmability of SDN makes it adaptable to the evolving requirements of vehicular networks. However, the centralized control inherent in SDN introduces new vulnerabilities, such as controller attacks. Additionally, managing the complexity

of SDVN systems while ensuring both security and privacy is a critical issue that must be addressed.

B. TRENDS IN PSEUDONYM LIFE CYCLE IMPLEMENTATION

In VANETs, the pseudonym life cycle serves as a critical mechanism for maintaining vehicle privacy and security. This life cycle involves the management of pseudonyms, which act as temporary identifiers that allow vehicles to communicate anonymously while ensuring secure and authenticated interactions. We analyzed whether the reviewed research studies fully implemented the pseudonym life cycle, which consists of five stages: issuance, usage, changes, resolution, and revocation. Table 9 presents the results of the extent of pseudonym life cycle implementation across the examined studies.

1) ISSUANCE

Pseudonym issuance is the initial stage of the pseudonym life cycle. The primary purpose of pseudonym issuance is to protect the privacy of vehicles by ensuring that their true identities remain concealed while still enabling secure communication and authentication within the network. For this stage, we identified the entities responsible for pseudonym generation, the methods employed, and the inputs utilized in the pseudonym generation process. Pseudonym issuance in VANETs can be performed either by individual entities or through collaboration between multiple entities. Our analysis revealed that the majority of pseudonym generation is carried out by vehicles/OBUs and TAs, accounting for 37% and 27% of PR studies, respectively. Additionally, pseudonym generation can be handled jointly by two entities, such as the vehicle and the TA, the vehicle and the RSU, the vehicle and the TRA, the vehicle and the KGC, and the vehicle and the RTA, collectively accounting for 23% studies. Among these collaborative approaches, the most common combinations in PR studies involve vehicles and TAs (46%) and vehicles and TRAs (35%). Other combinations, such as RSU-TA and RSU-CA, were less frequently observed, together representing only 2% of PR studies. Table 10 provides a detailed overview of the various entities responsible for pseudonym generation across the primary research papers.

With respect to the techniques used for pseudonym generation, the majority of the research papers, approximately

TABLE 9. Pseudonym lifecycle and associated primary research (PR) sources.

Pseudonym lifecycle					Sources	Total
Issuance	Usage	Changing	Resolution	Revocation		
✓	✓	✓	✓	✓	PR01 [13], PR04 [15], PR06 [17], PR07 [119], PR08 [18], PR14 [22], PR27 [95], PR47 [45], PR49 [47], PR73 [66], PR76 [69], PR77 [70], PR93 [80], PR111 [107]	14
✓	✓	✓	✓	-	PR03 [14], PR24 [120], PR26 [30], PR33 [36], PR44 [42], PR54 [110], PR86 [77]	7
✓	✓	-	✓	✓	PR09 [19], PR12 [109], PR13 [21], PR17 [25], PR21 [28], PR23 [29], PR28 [31], PR36 [97], PR38 [38], PR40 [98], PR42 [41], PR56 [53], PR60 [122], PR61 [115], PR64 [58], PR65 [59], PR68 [61], PR72 [65], PR74 [67], PR75 [68], PR80 [100], PR81 [73], PR82 [74], PR83 [75], PR84 [76], PR87 [101], PR91 [102], PR92 [123], PR99 [103], PR100 [84], PR102 [104], PR104 [105], PR105 [87], PR106 [106], PR113 [92]	35
✓	✓	✓	-	-	PR16 [24], PR25 [114], PR52 [50]	3
✓	✓	-	✓	-	PR05 [16], PR10 [20], PR15 [23], PR19 [26], PR29 [32], PR30 [33], PR34 [120], PR37 [37], PR39 [39], PR41 [40], PR43 [121], PR45 [43], PR46 [44], PR48 [46], PR50 [48], PR51 [49], PR53 [51], PR55 [52], PR57 [54], PR59 [111], PR66 [99], PR67 [60], PR69 [62], PR70 [63], PR71 [64], PR78 [71], PR79 [72], PR89 [116], PR90 [79], PR94 [81], PR96 [83], PR97 [117], PR98 [118], PR103 [86], PR107 [88], PR108 [89], PR109 [113], PR112 [91]	38
✓	✓	-	-	-	PR02 [108], PR11 [124], PR18 [125], PR20 [27], PR22 [93], PR31 [34], PR32 [35], PR35 [96], PR58 [55], PR62 [56], PR63 [57], PR85 [112], PR88 [78], PR95 [81], PR101 [85], PR110 [90]	16

TABLE 10. Entities that generate pseudonyms and associated primary research (PR) sources.

Entity	Source	Total	Freq.
Vehicle/OBU	PR01 [13], PR02 [108], PR07 [119], PR10 [20], PR12 [109], PR13 [21], PR15 [23], PR19 [26], PR20 [27], PR21 [28], PR23 [29], PR25 [114], PR26 [30], PR28 [31], PR29 [32], PR30 [33], PR33 [36], PR35 [96], PR40 [98], PR42 [41], PR48 [46], PR49 [47], PR56 [53], PR57 [54], PR61 [115], PR62 [56], PR64 [58], PR65 [59], PR66 [99], PR68 [61], PR72 [65], PR75 [68], PR82 [74], PR83 [75], PR84 [76], PR85 [112], PR91 [102], PR95 [82], PR97 [117], PR98 [118], PR99 [103], PR106 [106]	42	42/113
TA	PR03 [14], PR04 [15], PR05 [16], PR17 [25], PR18 [125], PR22 [93], PR24 [94], PR31 [34], PR34 [120], PR36 [97], PR38 [38], PR44 [42], PR45 [43], PR46 [44], PR51 [49], PR60 [122], PR63 [57], PR70 [63], PR80 [100], PR81 [73], PR88 [78], PR89 [116], PR90 [79], PR92 [123], PR93 [80], PR101 [85], PR103 [86], PR104 [105], PR105 [87], PR107 [88], PR111 [107]	30	30/113
CA	PR11 [124], PR54 [110], PR58 [55], PR59 [111], PR100 [84]	5	5/113
RSU	PR06 [17], PR09 [19], PR74 [67]	3	3/113
TRA	PR41 [40], PR50 [48]	2	2/113
SM	PR27 [95]	1	1/113
KGC	PR39 [39]	1	1/113
Vehicle and TA	PR08 [18], PR55 [52], PR71 [64], PR73 [66], PR76 [69], PR77 [70], PR79 [72], PR86 [77], PR96 [83], PR108 [89], PR109 [113], PR113 [92]	12	12/113
Vehicle and TRA	PR32 [35], PR37 [37], PR43 [121], PR53 [51], PR67 [60], PR78 [71], PR87 [101], PR110 [90], PR112 [91]	9	9/113
Vehicle and KGC	PR69 [62], PR102 [104]	2	2/113
Vehicle and RSU	PR14 [22]	1	1/113
Vehicle and Regional TA	PR52 [50]	1	1/113
Vehicle and CDA	PR94 [81]	1	1/113
RSU and TA	PR16 [24]	1	1/113
RSU dan CA	PR47 [45]	1	1/113

73%, implement hash functions. Hash functions produce fixed-length outputs from input data (e.g., vehicle identity),

which are designed to be difficult to reverse engineer. This characteristic ensures that the real identity of a vehicle

remains obfuscated, thereby strengthening privacy protection. Additionally, since VANETs often operate in resource-constrained environments, hash functions are particularly suitable because of their low computational overhead, making them ideal for vehicles with limited processing power, such as OBUs. Other techniques used for pseudonym generation include encryption and digital signatures. Regarding the input used for pseudonym generation, nearly 90% of the PR studies identified the original vehicle's ID as a key input. In addition, several studies incorporated a validity timestamp for the pseudonym. This timestamp facilitates smoother pseudonym changes, ensuring that pseudonyms are updated at appropriate intervals while maintaining privacy and security.

2) USAGE

After issuance, a vehicle utilizes its pseudonym for secure communication within a VANET, ensuring that its true identity remains protected. This pseudonym is employed during both V2V and V2I communications to maintain anonymity during essential network operations, such as sharing traffic information, transmitting safety warnings, and engaging in cooperative driving tasks. By using pseudonyms, vehicles can interact with other entities in a network without compromising the privacy of their original identities, which is crucial for preserving user confidentiality in high-mobility environments such as VANETs. All the reviewed studies used pseudonyms in their communication schemes.

3) CHANGING

To prevent long-term tracking of vehicles by malicious entities, pseudonyms are periodically changed. This process ensures that vehicles cannot be easily linked across different time frames or locations and thus enhances privacy by minimizing the chances of correlating multiple pseudonyms to a single vehicle. In 21% of the research papers reviewed, pseudonym changing mechanisms were implemented. According to ETSI TR 103415 [12], six pseudonym changing strategies are identified: fixed parameter, randomness, silent period, vehicle-centric, density-based, and mix zone. We identified the specific pseudonym changing strategies used in these studies, as shown in Table 11. This analysis highlights the diversity of approaches that researchers are implementing to ensure that vehicles can regularly change their pseudonyms to maintain privacy while preserving network functionality.

4) RESOLUTION

In scenarios where a vehicle is involved in illegal activities or traffic incidents, the resolution of its pseudonym becomes crucial for identifying the vehicle's true identity. This process is typically executed by authorized entities, such as law enforcement agencies, which utilize cryptographic information securely maintained by the TA or TRA. The resolution mechanism is structured to be a controlled and transparent process, ensuring vehicle privacy while

TABLE 11. Pseudonym changing strategy and associated primary research (PR) sources.

Strategies	Sources	Total	Freq.
Vehicle-centric	PR01 [13], PR07 [119], PR25 [114], PR26 [30], PR27 [95], PR33 [36], PR47 [45], PR49 [47]	8	8/24
Fixed parameter	PR08 [18], PR52 [50], PR76 [69], PR77 [70], PR86 [77], PR93 [80], PR111 [107]	7	7/24
Randomness	PR16 [24]	1	1/24
Vehicle-centric and randomness	PR03 [14], PR14 [22]	2	2/24
Vehicle-centric and fixed parameter	PR04 [15]	1	1/24
Others	PR06 [17], PR44 [115], PR73 [66], PR24 [94], PR54 [110]	5	5/24

simultaneously facilitating accountability and traceability. During resolution, the TA or TRA acts as a mediator, ensuring that sensitive information is disclosed only to entities with legitimate authority and a valid reason for obtaining the vehicle's identity. This careful balance between privacy and accountability is paramount, especially in a context where vehicles continuously communicate within VANETs. Moreover, the resolution process is designed to minimize the risk of abuse or unauthorized access to vehicle identities, employing robust security protocols and access controls to safeguard data integrity. The overarching goal is to establish a framework in which privacy is respected, yet law enforcement and regulatory bodies can effectively investigate and respond to incidents. This dual focus on privacy preservation and accountability is essential for maintaining public trust in VANET systems, ensuring that users feel secure in their interactions while being aware that mechanisms exist to address misconduct when necessary. Thus, the pseudonym resolution process plays a critical role in sustaining the operational integrity and societal acceptance of VANET technologies. A total of 83% of the reviewed research papers addressed resolution in the pseudonym life cycle.

5) REVOCATION

If a vehicle is determined to be malicious or has violated network policies, its pseudonym can be revoked. This step is crucial for ensuring that the vehicle is no longer able to communicate within the network via its current pseudonym, thus effectively removing it from the system and preventing further interactions that could compromise network safety. The revocation process typically involves authorized entities, such as the TA, which issues a revocation notice on the basis of validated evidence of the vehicle's misconduct. This measure not only safeguards the network but also enhances trust among legitimate users by reinforcing accountability. Revocation is essential for maintaining the integrity and security of the network, as it acts as a deterrent against

TABLE 12. Security and privacy analysis and associated primary research (PR) sources.

Analysis	Sources	Total	Freq.
Formal and Informal	PR01 [13], PR02 [108], PR03 [14], PR04 [15], PR05 [16], PR07 [119], PR09 [19], PR12 [109], PR13 [21], PR14 [22], PR15 [23], PR16 [24], PR17 [25], PR18 [125], PR19 [26], PR20 [27], PR21 [28], PR23 [29], PR24 [94], PR25 [114], PR29 [32], PR30 [33], PR31 [34], PR32 [35], PR33 [36], PR34 [120], PR35 [96], PR37 [37], PR38 [38], PR39 [39], PR40 [98], PR41 [40], PR43 [121], PR44 [42], PR46 [44], PR47 [45], PR48 [46], PR49 [47], PR50 [48], PR51 [49], PR52 [50], PR53 [51], PR55 [52], PR56 [53], PR61 [115], PR62 [56], PR64 [58], PR67 [60], PR68 [61], PR69 [62], PR71 [64], PR72 [65], PR73 [66], PR74 [67], PR75 [68], PR76 [69], PR77 [70], PR78 [71], PR79 [72], PR81 [73], PR82 [74], PR83 [75], PR84 [76], PR85 [112], PR86 [77], PR87 [101], PR90 [79], PR91 [102], PR93 [80], PR94 [81], PR95 [82], PR97 [117], PR98 [118], PR101 [85], PR102 [104], PR105 [87], PR106 [106], PR107 [88], PR108 [89], PR110 [90], PR111 [107], PR112 [91], PR113 [92]	83	83/113
Formal only	PR26 [30], PR54 [110], PR57 [54], PR59 [111], PR60 [122], PR70 [63], PR96 [83], PR109 [113]	8	8/113
Informal only	PR08 [18], PR10 [20], PR11 [124], PR22 [93], PR27 [95], PR28 [31], PR36 [97], PR42 [41], PR45 [43], PR58 [55], PR63 [57], PR65 [59], PR66 [99], PR80 [100], PR89 [116], PR92 [123], PR99 [103], PR100 [84], PR103 [86], PR104 [105]	20	20/113
N/A	PR06 [17], PR88 [78]	2	2/113

malicious behaviors and helps to exclude vehicles that pose a threat. Furthermore, the ability to rapidly revoke pseudonyms in response to detected threats is critical for adaptive security mechanisms within VANETs, ensuring that the network can respond effectively to evolving risks while preserving user privacy and data confidentiality. A total of 43% of the research papers addressed resolution in the pseudonym lifecycle.

C. PRIVACY ANALYSIS TREND

Privacy analysis in VANETs is conducted to evaluate the effectiveness of privacy-preserving schemes in protecting VANET entities' data. This analysis focuses on assessing the mechanisms used to prevent unauthorized access to vehicle identities, locations, and communication data. Generally, various aspects are assessed, including the effectiveness of pseudonym schemes, cryptographic techniques, and data anonymity. We classified the privacy analysis conducted in the research papers into two types: formal and informal.

Formal analysis involves the use of mathematical models, proofs, or formal methods to assess the privacy guarantees of a system thoroughly. In contrast, informal analysis is based on empirical or heuristic methods, where privacy considerations are evaluated through simulations, experimental setups, or qualitative analysis, often without formal proofs. As shown in Table 12, the majority of research studies incorporated both formal and informal privacy analyses. However, 7% of the studies used only formal analysis, whereas 16% used only informal analysis. Only two primary research studies did not conduct a privacy-preserving analysis.

We conducted a deeper investigation into the formal analysis performed in the primary research studies. We found that the formal analysis addressed not only privacy but also security, as privacy-preserving schemes are often integrated with security-preserving mechanisms. The dominant form of formal analysis involved security and privacy proofs of existing schemes. Several theoretical frameworks, such as the random oracle model (ROM), elliptic curve discrete logarithm problem (ECDLP), and elliptic curve Diffie–Hellman (ECDH), are frequently used to assess the security of cryptographic protocols. However, these frameworks are not specifically designed for formal privacy analysis. Nevertheless, in some cases, they indirectly support privacy analysis by evaluating cryptographic schemes that support privacy-preserving mechanisms. In addition to theoretical frameworks, tools such as BAN logic, the AVISPA, and SPAN are not explicitly designed for formal privacy analysis. Nevertheless, some studies have used tools commonly employed for formal privacy analysis, such as ProVerif, to analyze privacy properties such as anonymity and unlinkability. ProVerif was used in PR03 [14] and PR16 [24], which also utilized BAN logic to analyze the security of existing schemes. The mathematical models and tools used for both security and privacy analysis are detailed in Table 13.

IV. RESEARCH CHALLENGES AND FUTURE RESEARCH DIRECTIONS

The research challenges and potential paths for implementing pseudonyms to protect privacy in VANETs are described in this section. We also answer RQ4 in Section IV. As indicated by Table 9, only 12% of the reviewed studies fully implemented the pseudonym life cycle. Table 14 presents the relationships among the VANET system models used, the implementation of the pseudonym life cycle, and the privacy analysis performed in complete pseudonym life cycle implementations presented in PR studies.

A. SPoF IN VANET STANDARD SYSTEM MODEL

The preference for using standard system models in VANETs that appoint the TA as a central entity responsible for critical functions leads to the challenge of the SPoF. This problem can be addressed by distributing the critical functions of the TA to other trusted entities to help the TA perform these functions. In addition, implementing emerging technologies such as blockchain and fog/edge computing

TABLE 13. Formal analysis and associated primary research (PR) sources.

Formal Analysis	Source	Total	Freq.
Proof of security and privacy scheme	PR02 [108], PR09 [19], PR19 [26], PR20 [27], PR21 [28], PR24 [94], PR25 [114], PR26 [30], PR41 [40], PR46 [44], PR49 [47], PR54 [110], PR57 [54], PR60 [122], PR64 [58], PR67 [60], PR68 [61], PR69 [62], PR70 [63], PR71 [64], PR72 [65], PR75 [68], PR78 [71], PR91 [102]	24	24/91
ROM	PR01 [13], PR05 [16], PR07 [119], PR23 [29], PR29 [32], PR33 [36], PR34 [120], PR37 [37], PR38 [38], PR44 [42], PR73 [66], PR77 [70], PR87 [101], PR90 [79], PR93 [80], PR95 [82], PR96 [83], PR98 [118], PR106 [106], PR109 [113], PR112 [91]	21	21/91
ECDLP Assumption	PR31 [34], PR40 [98], PR51 [49], PR55 [52], PR59 [111], PR61 [115], PR82 [74], PR86 [77], PR105 [87], PR108 [89], PR111 [107], PR113 [92]	12	12/91
Under attack scenario	PR14 [22], PR32 [35], PR47 [45], PR56 [53], PR74 [67], PR79 [72], PR94 [81], PR102 [104], PR107 [88], PR110 [90]	10	10/91
BAN Logic	PR17 [25], PR62 [56], PR81 [73], PR83 [75], PR84 [76]	5	5/91
AVISPA	PR04 [15], PR12 [109]	2	2/91
Real or Random (RoR)	PR18 [125]	1	1/91
Under zero knowledge proof	PR35 [96]	1	1/91
DLP assumption	PR50 [48]	1	1/91
BAN logic and ProVerif	PR03 [14], PR16 [24]	2	2/91
BAN logic and ROM	PR13 [21], PR76 [69]	2	2/91
BAN logic and AVISPA	PR15 [23]	1	1/91
BAN logic, SPAN and AVISPA	PR52 [50]	1	1/91
ROM and ECDLP	PR30 [33], PR39 [39], PR43 [121]	3	3/91
ROM and AVISPA	PR43 [121]	1	1/91
ROM and Schyther Tool	PR101 [85]	1	1/91
AVISPA and RoR	PR85 [112], PR97 [117]	2	2/91
ECDLP and ECDH	PR53 [51]	1	1/91

can decentralize or distribute some TA functions. Future studies should investigate hybrid system architectures that integrate traditional models with contemporary technologies, including blockchain and cloud-based solutions. Such hybrid systems aim to leverage the benefits of decentralization and the computational efficiency provided by these innovative technologies.

B. POTENTIAL VULNERABILITIES RESULTING FROM AN INCOMPLETE PSEUDONYM LIFE CYCLE

The use of pseudonyms to maintain data privacy in VANETs needs to be implemented with a complete life cycle. Each stage in the life cycle, including issuance, usage, changing, revocation and resolution, plays a vital role in protecting the network from threats, ensuring vehicle privacy, and complying with applicable regulations. An incomplete implementation leaves a system vulnerable to attacks and privacy violations. Researchers should prioritize the development of a complete pseudonym life cycle while ensuring that performance is not compromised and that communication latency is minimized.

C. ROBUSTNESS TRADE-OFF AMONG VARIOUS PSEUDONYM ISSUANCE METHODS

One of the challenges of maintaining entity privacy in VANETs using pseudonyms is how to generate pseudonyms via the proper method. There will be a trade-off between the complexity of the function used and the resources and computing time needed. The methods used include the hash function, XOR function, signature, multiplication and encryption. Methods with a high level of complexity offer high security but have high computing resource requirements and longer processing times. In choosing the proper method to generate pseudonyms, future studies may consider anticipating the threat of quantum computing. As attackers computing power increases, the current use of hash functions may not be suitable for long-term use.

D. PSEUDONYM PREDICTABILITY

Predictable pseudonyms can lead to privacy vulnerabilities such as unauthorized tracking or linking of vehicle movements. The extent of this risk depends on how pseudonyms are generated. Currently, pseudonym generation often relies

TABLE 14. Complete pseudonym life cycle implementation and associated primary research (PR) sources.

PR	System model	Pseudonym life cycle				Privacy analysis type
		Issuance			Changing	
		Input	Method	Entity		
PR01 [13]	Standard	Signature of real ID	Converting signature	Vehicle	Vehicle-centric	Informal, formal: ROM
PR04 [15]	Standard	Vehicle ID, secret key (shared between vehicle and RSU), random number	Hash function	TA	Fixed parameter and vehicle-centric	Informal, formal: AVISPA
PR06 [17]	Standard	Pseudonym root, indicator level	Hash function	RSU	Pseudonym root	N/A
PR07 [119]	Blockchain and cloud-based	Vehicle's real ID, hashchain, timeslot	Hash function	Vehicle	Vehicle-centric	Informal, formal: ROM
PR08 [18]	Standard	Pseudo init: Vehicle ID, random number PID: Timestamp, group key, pseudo init	Hash function Hash function, concatenation, XOR	TA OBU	Fixed-parameter	Informal
PR14 [22]	Standard	PIDv: Vehicle ID, random number Proof: Vehicle ID, random number Ps: Proof, level	Hash function, XOR Hash function, concatenation Hash function, concatenation	OBU OBU RSU	Randomness Vehicle-centric	Informal Formal: Under attack scenario
PR27 [95]	Blockchain-based	SM secret key, vehicle's public key, timestamp	Hash function, concatenation	Service manager (SM)	Vehicle-centric	Informal
PR47 [45]	Standard	Primary pseudonym: expiration time, vehicle's public key Secondary pseudonym: Expiration time, new vehicle's public key	Encryption Concatenation	CA RSU	Vehicle-centric Vehicle-centric	Informal Formal: Under attack scenario
PR49 [47]	Standard	Pseudonym in V2I authentication: Random number, time parameter Pseudonym in V2V authentication: Pseudonym generation parameter	Multiplication	Vehicle	Vehicle-centric	Informal Formal: Security model proof
PR73 [66]	Standard	Pseudonym: Vehicle's real ID, validity period Pseudo ID: Random integer, public parameter, pseudonym, TA's public key	Hash function, concatenation Multiplication, XOR, hash function	TA OBU	Fixed parameter	Informal Formal: ROM
PR76 [69]	Standard	Pseudonym of vehicle: Vehicle's identity, sort period Pseudo ID: Pseudonym, random number, TA's public key	Hash function, concatenation Multiplication, XOR, hash function	TA OBU	Fixed parameter	Informal Formal: ROM and BAN Logic
PR77 [70]	Standard	Pseudonym of vehicle: Vehicle's identity, validity period Pseudo ID: Pseudonym, random number, TA's public key	Hash function, concatenation Multiplication, XOR, hash function	TA OBU	Fixed parameter	Informal Formal: ROM
PR93 [80]	Standard	Random number, generator group G, TA's public key, vehicle ID, validity period of pseudo identity	Multiplication, XOR, hash function	TA	Fixed parameter	Informal, Formal: ROM
PR111 [107]	Blockchain	Vehicle ID, system secret key, effective period of current pseudonym vehicle ID, secret key (shared between vehicle and RSU), random number	hash function, XOR, concatenation	TA	Fixed parameter	Informal, Formal: ECDLP

on a fixed combination of inputs in the form of the vehicle's real identity, time stamp, and random number, which makes it easier for attackers to track or link pseudonyms over time, especially if the pseudonym change interval is predictable. To overcome this issue, researchers need to add dynamic inputs such as road conditions, vehicle routes, and geographical locations. However, an important aspect to consider in future studies is balancing unpredictability with traceability.

E. DIVERSITY OF PSEUDONYM CHANGING STRATEGIES

Pseudonym-changing strategies vary significantly across different approaches. These strategies can be vehicle-centric, fixed parameter, random, or a combination of multiple techniques. This diversity of tactics highlights the difficulties of developing a single-solution approach that can be implemented uniformly across all scenarios and environments, regardless of variables such as vehicle density, speed, or network conditions. The variety of solutions presented in the analyzed publications indicates that such a general approach has not been developed. Instead, diverse conditions may necessitate adapted or adaptable methods to provide optimal privacy and security, which privacy-preserving scheme designers must consider.

F. INFORMAL VS. FORMAL ANALYSIS

Current privacy analyses in VANET research predominantly rely on informal evaluation methods, which may lead to potential privacy vulnerabilities due to the lack of precision in the qualitative analysis of privacy requirements. In terms of formal analysis, more emphasis is currently placed on security analysis. Few studies have conducted formal analyses using tools designed to analyze privacy. In future studies, although privacy-preserving schemes are often integrated with security-preserving schemes, researchers should also focus on implementing formal analysis for privacy-preserving schemes.

V. CONCLUSION

The purpose of the current study was to analyze research trends, challenges, and future directions in the privacy-preserving schemes of VANETs by identifying the VANET system models used, the implementation of the pseudonym life cycle, and the privacy analysis of the privacy-preserving schemes applied in the research papers. The results showed that 71% of the reviewed studies used standard system models, 12% implemented a comprehensive pseudonym life cycle, and 73% conducted formal and informal analyses. Thereafter, we identified challenges and future directions for research, focusing on the following areas: overcoming an SPoF in the standard VANET model, potential vulnerabilities due to incomplete pseudonym life cycle implementation, robustness trade-offs between various pseudonym issuance methods, pseudonym predictability, appropriate pseudonym change strategies, and robust privacy analysis. We hope that this paper provides

a clearer understanding of the privacy preservation trend in VANETs from the perspective of the VANET system model, pseudonym life cycle implementation, and privacy analysis conducted by researchers. Additionally, researchers can consider these findings when developing a robust privacy preservation scheme for VANETs.

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